Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/23/2006
PI Name:	Civitelli, Roberto M.D.		
Project Title:	Osteoblast Regulation by Mechanical Stimuli via Gap Junctions		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63110-1093	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-02
Start Date:	07/01/2000	**End Date:**	12/31/2003
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1412		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Specific Aim 1: Define the molecular mechanisms by which mechanical stimuli regulate gap junctional communication between osteoblasts. Specific Aim 2: Determine the gap junctional communication requirements for mechanical regulation of osteoblast function. The hypothesis is that the type of gap junctional communication provided by connex43 (Cx43) is required for mechanotransduction in osteoblasts. Interference with Cx43-mediated gap junctional communication disrupts osteoblast function and response to mechanical stimuli.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Osteoblasts form a tightly networked, functional syncytium via gap junctional intercellular communication and respond to hormonal, chemical and physcial stimuli as a multicellular unit. The work that is being performed under this project will unlock a mechanism that that may explain the reduced bone formation and bone loss in conditions of unloading, such as in space flights.
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)